TECHNICAL SUPPLEMENT 07 · ENGLISH

State of the Art and Technical References

Agri-Social Navigation in Vineyard Corridors

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Study identity	Specification
Campaign / Campaña	EVAL_20261019_112257
Navigation / Navegación	9 × 4 × 10 = 360; SUCCESS 266; TIMEOUT 94
Benchmark	180; 20 per method / por método
Version / Versión	20 October / octubre 2026

Numerical companion to the revised Sensors manuscript. This report uses the identified endpoint dataset and separates it from the older public campaigns. All seven authors are listed in the original manuscript order.

1. Literature and implementation scope

Agricultural HRI motivates shared-workspace navigation. Vasconez, Kantor and Auat Cheein survey agricultural collaboration; Vasconez, Guevara and Auat Cheein study visual non-verbal communication and approaching/following/avoiding workers in an avocado-harvesting simulation. These works motivate context, not numerical validation of the present results.

M4 uses a longitudinal-lateral anisotropic field with front-rear symmetry and nearest-human selection. B2 is ORCA-inspired and does not implement the reference half-plane linear program; B4 is barrier-inspired sampled control and does not inherit a continuous-time CBF safety proof. General social-navigation surveys and mathematical reference algorithms must be distinguished from these evaluated implementations.

2. Manuscript bibliography

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Vasconez.pdf and six reference PDFs were recovered unchanged from the preceding repository revision. The bibliography preserves their authorship. Listing the study authors here does not change the authorship of cited publications.

Source and scope

Source: data/; results/expected/. PDFs provide the reading reports; CSV/JSON retain numerical inputs and provenance. Scope: evaluated simulation, without new experiments, independent raw-stream reconstruction or physical validation.